Hurdle-Lognormal state-space model

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Abstract

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1. Introduction

Many insurance payment variables have a mixed distribution consisting of a point mass at zero and a continuous distribution over the positive real line. A zero observation may represent the absence of an insured event, a claim, or a payment during a given period, whereas a positive observation records the corresponding payment amount. Such data can be modelled through several approaches, including collective risk models, Tweedie distributions, and two-part models. In a two-part model, the loss at time t is represented as

$$Y_t = Z_t S_t,$$

where the binary variable Z_t indicates whether a positive loss occurs and the positive continuous variable S_t represents its amount. This decomposition makes it possible to model the occurrence and positive amount separately while retaining a direct interpretation of their combined effect on the total loss.

For longitudinal ratemaking, it is important to account for dependence among repeated losses from the same policyholder or risk. Such dependence is commonly introduced through random effects representing unobserved heterogeneity. In a two-part model, a shared random effect may simultaneously influence the occurrence and positive amount, or separate but correlated random effects may be assigned to the two components. The latter formulation is more flexible because it allows the unobserved factors affecting occurrence and severity to differ, while their association is captured through the correlation between the random effects. This flexibility, however, generally makes likelihood, posterior, and predictive calculations more difficult.

The distinction between tractability for estimation and tractability for prediction is particularly important in insurance. For parameter estimation, one seeks an analytically manageable likelihood function and posterior distribution of the random effects. For experience ratemaking, the principal quantity is instead the conditional expectation

$$\mathbb{E}\left[Y_{t+1} \mid Y_{1:t} = y_{1:t}\right],$$

which provides the one-step-ahead a posteriori premium under the equivalence principle. For excess-of-loss coverage, the related conditional stop-loss expectation

$$\mathbb{E}\left[(Y_{t+1} - d)_+ \mid Y_{1:t} = y_{1:t}\right], \qquad d > 0,$$

is also of direct actuarial interest. A model may admit a tractable likelihood without providing corresponding analytical expressions for these predictive quantities. In many random-effects and copula-based two-part models, they must therefore be evaluated through additional numerical integration or simulation.

A particularly useful analytical structure arises from combining a Gaussian random effect with a probit link. For dependent binary observations, the resulting Bernoulli-probit random effect model expresses joint probabilities, posterior densities, and predictive probabilities through multivariate Gaussian cumulative distribution functions (Durante, 2019; Fasano et al., 2021; Cao et al., 2022). Two-part random-effects and stochastic-process models for longitudinal semicontinuous data have also been studied extensively in the statistical and biostatistical literature (Farewell et al., 2017; Yiu and Tom, 2018). In the actuarial literature, joint modelling of dependent binary insurance outcomes is likewise well established (Young et al., 2009; Frees et al., 2010; Guo et al., 2025; Oh et al., 2026). Nevertheless, the analytical advantages of the Gaussian-probit construction appear to have received limited attention for insurance two-part models, particularly for deriving conditional premium formulas under correlated occurrence and severity random effects.

Motivated by this gap, this paper studies a Bernoulli-lognormal probit two-part model with a bivariate Gaussian random effect. The first random effect enters the probit model for the occurrence indicator, whereas the second enters the lognormal model for the positive loss amount. Their correlation links the occurrence and severity components and allows positive-loss observations to update the latent risk associated with future claim occurrence. Thus, the model retains the interpretability and flexibility of the standard bivariate random-effects formulation while preserving a tractable Gaussian structure.

The main contribution of the paper is to show that this structure provides a fully analytical environment for both ratemaking and statistical inference. First, we derive a closed-form expression for

$$\mathbb{E}\left[Y_{t+1} \mid Y_{1:t} = y_{1:t}\right]$$

in terms of a ratio of multivariate Gaussian cumulative distribution functions. The derivation combines Gaussian updating based on the observed positive losses, Gaussian exponential tilting, and a Gaussian product-probit identity. Second, we derive a closed-form expression for the conditional stop-loss expectation

$$\mathbb{E}\left[(Y_{t+1} - d)_+ \mid Y_{1:t} = y_{1:t}\right],$$

which provides an analytical one-step-ahead premium for the layer above a deductible d. These formulas avoid direct numerical integration over the bivariate random effect.

For statistical inference, we further derive the mixed discrete-continuous likelihood of the observed loss history and the posterior density of the bivariate random effect. Consequently, the same Gaussian representation that produces the predictive premiums also provides tractable likelihood-based and Bayesian inference. To establish the main analytical mechanism, we first review the Bernoulli-probit random effect model and its closed-form joint probabilities, predictive probabilities, and posterior density. We then introduce the Bernoulli-lognormal bivariate random effect model and derive its one-step-ahead mean and stop-loss predictions, followed by the corresponding

likelihood and posterior results.

2. Survival representation of counting distributions

sec.3

Hurdle representations provides the flexibility in modelling the count data by decomposing a count distribution according to whether the count crosses a specified threshold. The standard hurdle representation uses the threshold, separating $Y = 0$ from $Y \geq 1$. It can be extended to a double-hurdle representation, separating $Y = 1$ from $Y \geq 2$ conditional on having already crossed the first threshold. Motivated by these models, this section presents a more general threshold-by-threshold view of count distributions, in which the count is described by its successive passages from one level to the next.

We begin with the notation used throughout the paper. Let $\mathbb{N}$ denote the set of positive integers and let

$$\mathbb{N}_0 := \mathbb{N} \cup \{0\}.$$

We write $\mathbb{R}$, $\mathbb{R}_{\geq 0}$, and $\mathbb{R}_{>0}$ for the real line, the nonnegative real line, and the positive real line, respectively. For $k \in \mathbb{N}$ and for any set A, we write A^k for the k-fold Cartesian product of A. In particular, $\mathbb{N}^k$, $\mathbb{N}_0^k$, $\mathbb{R}^k$, $\mathbb{R}_{\geq 0}^k$, and $\mathbb{R}_{>0}^k$ denote the corresponding k-dimensional product spaces.

For $k \in \mathbb{N}$, and for real numbers $x_1, \ldots, x_r$, we use

$$x_{1:k} := (x_1, \ldots, x_k)^\top \in \mathbb{R}^k$$

and all the vector is regarded as column vector. Especially, we use $0_{1:k} = (0, \cdots, 0)^\top \in \mathbb{R}^k$, and similarly, in a matrix environment, we write $0_{i\times j}$ to be $i \times j$ matrix of zero. We also write

$$e_{1:k} = (0, \cdots, 0, 1)^\top \in \mathbb{R}^k.$$

If $x_{1:k}, y_{1:k} \in \mathbb{R}^k$, their Euclidean inner product is denoted by

$$\langle x_{1:k}, y_{1:k} \rangle := \sum_{j=1}^{k} x_j y_j.$$

Throughout, we adopt the standard conventions that a product over an empty index set equals 1, whereas a sum over an empty index set equals 0. That is, if $\mathcal{I} = \varnothing$, then

$$\prod_{j\in\mathcal{I}} a_j := 1, \qquad \sum_{j\in\mathcal{I}} a_j := 0.$$

We then introduce some distributions used in this paper.

- Gamma(a, b): gamma distribution with shape parameter $a \in \mathbb{R}_{>0}$ and rate parameter$b \in \mathbb{R}_{>0}$.

- Beta(a, b): beta distribution on $(0, 1)$ with parameters $a \in \mathbb{R}_{>0}$ and $b \in \mathbb{R}_{>0}$.
- Pois(λ): Poisson distribution with mean $\lambda \in \mathbb{R}_{>0}$.
- Geo(p): geometric distribution on $\mathbb{N}_0$ with parameter $p \in (0, 1)$ so that its mean is $p/(1-p)$. mean $\lambda \in \mathbb{R}_{>0}$ and variance $\lambda + \lambda^2/\Gamma$
- N(μ, b^2): normal distribution with mean $\mu \in \mathbb{R}_{>0}$ and variance $\sigma^2 \in \mathbb{R}_{>0}$.
- MVN$_k(\mu_{1:k}, \mathbf{\Sigma})$: multivariate normal distribution with $\mu_{1:k} \in \mathbb{R}^k$ and a positive definite covariance matrix $\mathbf{\Sigma} \in \mathbb{R}^{k \times k}$.

For a positive semidefinite covariance matrix $\mathbf{\Sigma} \in \mathbb{R}^{k \times k}$, let

$$\Phi_k(x_{1:k}\,;\,\mathbf{\Sigma}), \qquad x_{1:k} \in \mathbb{R}^k,$$

denote the cumulative distribution function at $x_{1:k}$ of

$$\mathrm{MVN}_k(0_{1:k}, \mathbf{\Sigma}).$$

That is, if

$$X_{1:k} \sim \mathrm{MVN}_k(0_{1:k}, \mathbf{\Sigma}),$$

then

$$\Phi_k(x_{1:k}\,;\,\mathbf{\Sigma}) := \mathbb{P}\left(X_1 \le x_1, \ldots, X_k \le x_k\right).$$

If $\mathbf{\Sigma}$ is positive definite, let

$$\phi_k(x_{1:k}\,;\,\mathbf{\Sigma}), \qquad x_{1:k} \in \mathbb{R}^k,$$

denote the corresponding probability density function with respect to the k-dimensional Lebesgue measure.

When the covariance argument is omitted, we use the standard multivariate normal distribution:

$$\phi_k(x_{1:k}) := \phi_k(x_{1:k}\,;\,\boldsymbol{I}_k), \qquad \Phi_k(x_{1:k}) := \Phi_k(x_{1:k}\,;\,\boldsymbol{I}_k),$$

where $\boldsymbol{I}_k$ is the $k \times k$ identity matrix. Thus, $\phi_k(x_{1:k})$ and $\Phi_k(x_{1:k})$ are, respectively, the probability density function and cumulative distribution function of

$$\mathrm{MVN}_k(0_{1:k}, \boldsymbol{I}_k).$$

When $k = 1$, for $x \in \mathbb{R}$, we simply write $\phi(x)$ and $\Phi(x)$, omitting the subscript $k = 1$.

3. Review

Here we provide the useful auxiliary result for this paper.

The following is the well-known result of Gaussian exponential tilting.

lem.4 **Lemma 1.** *Let*

$$X \sim N(\mu, v),$$

where $\mu \in \mathbb{R}$ and $v > 0$. For $h \in \mathbb{R}$, let

$$\widetilde{X} \sim N(\mu + vh, v).$$

Then, for every Borel-measurable function $g : \mathbb{R} \to \mathbb{R}$ satisfying

$$\mathbb{E}\left[|\exp(hX)g(X)|\right] < \infty,$$

we have

$$\mathbb{E}\left[\exp(hX)g(X)\right] = \exp\left(h\mu + \frac{h^2 v}{2}\right) \mathbb{E}\left[g(\widetilde{X})\right].$$

Proof. The integrability assumption gives

$$\begin{aligned}
\infty &> \mathbb{E}\left[|\exp(hX)g(X)|\right] \\
&= \int_{\mathbb{R}} \exp(hx)|g(x)| \frac{1}{\sqrt{2\pi v}} \exp\left(-\frac{(x-\mu)^2}{2v}\right) \mathrm{d}x \\
&= \exp\left(h\mu + \frac{h^2 v}{2}\right) \int_{\mathbb{R}} |g(x)| \frac{1}{\sqrt{2\pi v}} \exp\left(-\frac{(x-(\mu+vh))^2}{2v}\right) \mathrm{d}x \\
&= \exp\left(h\mu + \frac{h^2 v}{2}\right) \mathbb{E}\left[\left|g(\widetilde{X})\right|\right]
\end{aligned}$$

which ensures

$$\mathbb{E}\left[\left|g(\widetilde{X})\right|\right] < \infty.$$

Consequently, the same calculation with $|g|$ replaced by g is justified and gives

$$\mathbb{E}\left[\exp(hX)g(X)\right] = \exp\left(h\mu + \frac{h^2 v}{2}\right) \mathbb{E}\left[g(\widetilde{X})\right],$$

which proves the result. □

The following is the multivariate version of the Gaussian exponential tilting identity.

lem.3 **Lemma 2** (Multivariate Gaussian exponential tilting)**.** *Let*

$$X_{1:t} \sim \mathrm{MVN}_t(\mu_{1:t}, \mathbf{\Sigma}),$$

where $\mu_{1:t} \in \mathbb{R}^t$ and $\mathbf{\Sigma} \in \mathbb{R}^{t\times t}$ is positive definite covariance matrix. For $h_{1:t} \in \mathbb{R}^t$, let

$$\widetilde{X}_{1:t} \sim \mathrm{MVN}_t\left(\mu_{1:t} + \mathbf{\Sigma} h_{1:t}, \mathbf{\Sigma}\right).$$

Then, for every Borel-measurable function $g : \mathbb{R}^t \to \mathbb{R}$ satisfying

$$\mathbb{E}\left[\left|\exp\left(h_{1:t}^\top X_{1:t}\right) g(X_{1:t})\right|\right] < \infty,$$

we have

$$\begin{aligned}&\mathbb{E}\left[\exp\left(h_{1:t}^\top X_{1:t}\right) g(X_{1:t})\right] \\ &\quad = \exp\left(h_{1:t}^\top \mu_{1:t} + \frac{1}{2} h_{1:t}^\top \mathbf{\Sigma} h_{1:t}\right) \mathbb{E}\left[g(\widetilde{X}_{1:t})\right].\end{aligned}$$

Proof. The integrability assumption gives

$$\begin{aligned}
&\infty > \mathbb{E}\left[\left|\exp\left(h_{1:t}^\top X_{1:t}\right) g(X_{1:t})\right|\right] \\
&= \int_{\mathbb{R}^t} \exp\left(h_{1:t}^\top x_{1:t}\right) |g(x_{1:t})| \frac{1}{(2\pi)^{t/2}|\mathbf{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(x_{1:t} - \mu_{1:t})^\top \mathbf{\Sigma}^{-1}(x_{1:t} - \mu_{1:t})\right) \mathrm{d}x_{1:t} \\
&= \int_{\mathbb{R}^t} |g(x_{1:t})| \frac{1}{(2\pi)^{t/2}|\mathbf{\Sigma}|^{1/2}} \\
&\qquad \times \exp\left(h_{1:t}^\top \mu_{1:t} + \frac{1}{2} h_{1:t}^\top \mathbf{\Sigma} h_{1:t} - \frac{1}{2}\left(x_{1:t} - \mu_{1:t} - \mathbf{\Sigma} h_{1:t}\right)^\top \mathbf{\Sigma}^{-1}\left(x_{1:t} - \mu_{1:t} - \mathbf{\Sigma} h_{1:t}\right)\right) \mathrm{d}x_{1:t} \\
&= \exp\left(h_{1:t}^\top \mu_{1:t} + \frac{1}{2} h_{1:t}^\top \mathbf{\Sigma} h_{1:t}\right) \\
&\quad \times \int_{\mathbb{R}^t} |g(x_{1:t})| \frac{1}{(2\pi)^{t/2}|\mathbf{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}\left(x_{1:t} - \mu_{1:t} - \mathbf{\Sigma} h_{1:t}\right)^\top \mathbf{\Sigma}^{-1}\left(x_{1:t} - \mu_{1:t} - \mathbf{\Sigma} h_{1:t}\right)\right) \mathrm{d}x_{1:t} \\
&= \exp\left(h_{1:t}^\top \mu_{1:t} + \frac{1}{2} h_{1:t}^\top \mathbf{\Sigma} h_{1:t}\right) \mathbb{E}\left[\left|g(\widetilde{X}_{1:t})\right|\right]
\end{aligned}$$

which ensures

$$\mathbb{E}\left[\left|g(\widetilde{X}_{1:t})\right|\right] < \infty.$$

Consequently, the same calculation with $|g|$ replaced by g is justified and gives

$$\begin{aligned}&\mathbb{E}\left[\exp\left(h_{1:t}^\top X_{1:t}\right) g(X_{1:t})\right] \\ &\quad = \exp\left(h_{1:t}^\top \mu_{1:t} + \frac{1}{2} h_{1:t}^\top \mathbf{\Sigma} h_{1:t}\right) \mathbb{E}\left[g(\widetilde{X}_{1:t})\right],\end{aligned}$$

which proves the result. □

lem.5 **Lemma 3** (Gaussian product-probit identity)**.** *Let*

$$X_{1:t} = (X_1, \ldots, X_t)^\top$$

be a Gaussian vector with mean $\mu_{1:t} \in \mathbb{R}^t$ *and covariance matrix* $\boldsymbol{\Sigma} \in \mathbb{R}^{t \times t}$*. Then*

$$\mathbb{E}\left[\Phi_t(X_{1:t})\right] = \mathbb{E}\left[\prod_{j=1}^{t} \Phi(X_j)\right] = \Phi_t\left(\mu_{1:t}; \boldsymbol{I}_t + \boldsymbol{\Sigma}\right).$$

Proof. Let

$$W_{1:t} := (W_1, \ldots, W_t)^\top \sim \mathrm{MVN}(0_{1:t}, \boldsymbol{I}_t)$$

be independent of $X_{1:t}$. Then, we have

$$\begin{aligned}
\Phi_t(X_{1:t}) &= \mathbb{P}\left(W_{1:t} \leq X_{1:t} \mid X_{1:t}\right) \\
&= \prod_{j=1}^{t} \mathbb{P}\left(W_j \leq X_j \mid X_j\right) \\
&= \prod_{j=1}^{t} \Phi(X_j)
\end{aligned}$$

which further implies

$$\mathbb{E}\left[\Phi_t(X_{1:t})\right] = \mathbb{E}\left[\mathbb{E}\left[\prod_{j=1}^{t} \Phi(X_j)\right]\right].$$

Furthermore, from the following observations

$$\mathbb{E}\left[\Phi_t(X_{1:t})\right] = \quad = \mathbb{P}\left(W_{1:t} - (X_{1:t} - \mu_{1:t}) \leq \mu_{1:t}\right)$$

and

$$W_{1:t} - (X_{1:t} - \mu_{1:t}) \sim \mathrm{MVN}\left(0_{1:t}, \boldsymbol{I}_t + \boldsymbol{\Sigma}\right),$$

we conclude

$$\mathbb{P}\left(W_{1:t} - (X_{1:t} - \mu_{1:t}) \leq \mu_{1:t}\right) = \Phi_t\left(\mu_{1:t}; \boldsymbol{I}_t + \boldsymbol{\Sigma}\right),$$

which proves the result. □

3.1. Review: Bernoulli-probit random effect model

Joint modelling of dependent binary observations is an important topic in the statistical literature. Among the available approaches, models combining a Gaussian random effect with a probit link have recently attracted considerable attention because they flexibly accommodate dependence

among repeated binary observations while retaining analytically tractable Gaussian representations (Farewell et al., 2017; Durante, 2019; Fasano et al., 2021; Cao et al., 2022).

Although joint modelling of binary observations is also well established in the actuarial and insurance literature (Young et al., 2009; Frees et al., 2010; Guo et al., 2025; Oh et al., 2026), the analytical tractability offered specifically by the Bernoulli-probit random effect model appears to have received limited attention in actuarial applications. We therefore briefly review its random-effect formulation and derive closed-form expressions for the joint probability, one-step prediction, and posterior distribution of the random effect. Although these results are known in the statistical literature; see, for example, Farewell et al. (2017), we include them for completeness and as a foundation for the Bernoulli-lognormal models developed in the subsequent sections.

model.1 **Model 1** (Bernoulli-probit AR(1) state-space model)**.** *Let*

$$\sigma \in \mathbb{R}_{>0} \quad \textit{and} \quad (c_t)_{t\in\mathbb{N}} \subseteq \mathbb{R}$$

be predetermined exogenous constants. We define the sequence $(Z_t)_{t\in\mathbb{N}}$ *through the following random effect assumption and observation equation.*

i. Random effect: Assume

$$\Theta \sim \mathrm{N}(0, \sigma^2).$$

ii. Observation equation: Conditional on the random effect Θ*, assume that* $Z_1, Z_2, \ldots$ *are mutually independent and*

$$Z_t \mid \Theta \sim \mathrm{Ber}\left(\Phi(c_t + \Theta)\right), \qquad t = 1, 2, \ldots.$$

The following theorem provides closed-form expressions for the joint probability of the observed binary history and the corresponding one-step-ahead prediction under Model 1. In particular, the joint probability, and hence the likelihood, is expressed in terms of a multivariate normal cumulative distribution function, whereas the one-step-ahead prediction is obtained as the ratio of the joint probabilities for the augmented and observed histories.

thm.1 **Theorem 1** (Closed-form one-step prediction)**.** *Consider Model 1. For a realized history*

$$Z_{1:t} = z_{1:t}, \qquad \textit{for} \quad z_1, \cdots, z_t \in \{0, 1\},$$

define

$$x_j := \begin{cases} 2z_j - 1, & j = 1, \ldots, t; \\ 1, & j = t + 1. \end{cases} \tag{3.1}$$

eq.10

For $k \in \{t, t+1\}$, define the $k \times k$ matrix

$$\boldsymbol{X}_k := x_{1:k} x_{1:k}^\top = [x_i x_j]_{i,j=1}^k . \tag{3.2}$$ eq.8

Then, we have the following results.

i. The joint probability is obtained by

$$\mathbb{P}\left(Z_{1:t} = z_{1:t}\right) = \Phi_t\left((x_1 c_1, \ldots, x_t c_t)^\top ; \boldsymbol{I}_t + \sigma^2 \boldsymbol{X}_t\right).$$

ii. The one-step-ahead prediction is obtained by

$$\mathbb{E}\left[Z_{t+1} \mid Z_{1:t} = z_{1:t}\right] = \frac{\Phi_{t+1}\left((x_1 c_1, \ldots, x_t c_t, x_{t+1} c_{t+1})^\top ; \boldsymbol{I}_{t+1} + \sigma^2 \boldsymbol{X}_{t+1}\right)}{\Phi_t\left((x_1 c_1, \ldots, x_t c_t)^\top ; \boldsymbol{I}_t + \sigma^2 \boldsymbol{X}_t\right)}.$$

Proof. We first show part i.

$$\mathbb{P}\left(Z_{1:t} = z_{1:t}\right) = \Phi_t\left((x_1 c_1, \ldots, x_t c_t)^\top ; \boldsymbol{I}_t + \sigma^2 \boldsymbol{X}_t\right). \tag{3.3}$$ eq.thm1.his

By the conditional independence of $Z_1, \ldots, Z_t$ and the observation equation in Model 1,

$$\begin{aligned}
\mathbb{P}\left(Z_{1:t} = z_{1:t}\right) &= \mathbb{E}\left[\prod_{j=1}^t \mathbb{P}\left(Z_j = z_j \mid \Theta\right)\right] \\
&= \mathbb{E}\left[\prod_{j=1}^t \Phi\left(x_j(c_j + \Theta)\right)\right] \\
&= \Phi_t\left((x_1 c_1, \ldots, x_t c_t)^\top ; \boldsymbol{I}_t + \sigma^2 \boldsymbol{X}_t\right)
\end{aligned}$$

where the last equality follows from Lemma 3 and

$$(x_1(c_1 + \Theta), \cdots, x_t(c_t + \Theta))^\top \sim \mathrm{MVN}_t\left((x_1 c_1, \cdots, x_t c_t)^\top, \sigma^2 \boldsymbol{X}_t\right).$$

This proves (3.3).

Now, we prove part ii, by first observing

$$\begin{aligned}
\mathbb{E}\left[Z_{t+1} \mid Z_{1:t} = z_{1:t}\right] &= \mathbb{P}\left(Z_{t+1} = 1 \mid Z_{1:t} = z_{1:t}\right) \\
&= \frac{\mathbb{P}\left(Z_{1:t+1} = (z_1, \ldots, z_t, 1)^\top\right)}{\mathbb{P}\left(Z_{1:t} = z_{1:t}\right)}.
\end{aligned} \tag{3.4}$$ eq.1

where the denominator is in (3.3). Applying the result in part i to the numerator in (3.4), we obtain

$$\mathbb{P}\left(Z_{1:t+1} = (z_1, \ldots, z_t, 1)^\top\right) = \Phi_{t+1}\left((x_1c_1, \ldots, x_tc_t, x_{t+1}c_{t+1})^\top; \boldsymbol{I}_{t+1} + \sigma^2 \boldsymbol{X}_{t+1}\right)$$

which proves the result. □

The following corollary provides a closed-form expression for the posterior density of the random effect conditional on the observed binary history under Model 1. The result follows directly from Bayes' formula and the joint probability obtained in part i of Theorem 1.

cor.2 **Corollary 1** (Posterior distribution of the random effect)**.** *Under the setting of Theorem 1, the conditional distribution*

$$\Theta \mid z_{1:t} = z_{1:t}$$

has density $\pi(\theta \mid Z_{1:t} = z_{1:t})$ *at* $\theta \in \mathbb{R}$ *given by*

$$\pi(\theta \mid Z_{1:t} = z_{1:t}) = \frac{\frac{1}{\sigma}\phi\left(\frac{\theta}{\sigma}\right)\prod_{j=1}^{t}\Phi\left(x_j(c_j+\theta)\right)}{\Phi_t\left((x_1c_1, \ldots, x_tc_t)^\top; \boldsymbol{I}_t + \sigma^2\boldsymbol{X}_t\right)}. \tag{3.5}$$ eq.posterio

Proof. Let $\pi(\theta)$ denote the prior density of Θ at $\theta \in \mathbb{R}$. By Bayes' formula, we have

$$\begin{aligned}
\pi(\theta \mid Z_{1:t} = z_{1:t}) &= \frac{\pi(\theta)\mathbb{P}\left(Z_{1:t} = z_{1:t} \mid \Theta = \theta\right)}{\mathbb{P}\left(Z_{1:t} = z_{1:t}\right)} \\
&= \frac{\frac{1}{\sigma}\phi\left(\frac{\theta}{\sigma}\right)\prod_{j=1}^{t}\mathbb{P}\left(Z_j = z_j \mid \Theta = \theta\right)}{\mathbb{P}\left(Z_{1:t} = z_{1:t}\right)} \\
&= \frac{\frac{1}{\sigma}\phi\left(\frac{\theta}{\sigma}\right)\prod_{j=1}^{t}\Phi\{x_j(c_j+\theta)\}}{\mathbb{P}\left(Z_{1:t} = z_{1:t}\right)}.
\end{aligned}$$

Finally, substituting the expression for $\mathbb{P}\left(Z_{1:t} = z_{1:t}\right)$ from part i of Theorem 1 gives (3.5), which proves the result. □

4. Bernoulli-lognormal probit bivariate random effect model

sec.6

In the following we extend Model 1 additionally introduce positive part.

m.2.uni **Model 2** (Bernoulli-lognormal probit shared random effect model)**.** *Let*

$$\eta, \sigma, \tau > 0, \quad (c_t)_{t\in\mathbb{N}}, (d_t)_{t\in\mathbb{N}} \subseteq \mathbb{R}$$

be predetermined exogenous constants. We define the sequence $(Y_t)_{t\in\mathbb{N}}$ *through the following hierarchical specification.*

i. Random effect: Assume

$$\Theta \sim \mathrm{N}(0, \sigma^2).$$

ii. Observation equations: Conditional on Θ, assume that the pairs (Z_t, S_t), $t \in \mathbb{N}$, are mutually independent and that, for each t, their components are independent, with

$$Z_t \mid \Theta \sim \mathrm{Ber}\left(\Phi(c_t + \Theta)\right), \qquad S_t \mid \Theta \sim \mathrm{LN}(d_t + \eta\Theta, \tau^2).$$

The observed loss is defined by

$$Y_t = Z_t S_t, \qquad t \in \mathbb{N}.$$

Model 2 is equivalent to Model 3 with $\rho = 1$, under the identification

$$\Theta_1 = \Theta, \qquad \Theta_2 = \eta\Theta, \qquad \sigma_1 = \sigma, \qquad \sigma_2 = \eta\sigma.$$

The parameter η allows the shared random effect to enter the occurrence and log-severity components with different magnitudes.

The following remark collects the notation used in the results for Model 2.

rem.1 **Remark 1.** *Under Model 2, an observed history $Y_{1:t} = y_{1:t} \in \mathbb{R}^t_{\geq 0}$ reveals the information on all occurrence indicators, but only the severities corresponding to positive losses. For $j = 1, \ldots, t$, define*

$$z_j := 1_{[y_j > 0]}, \qquad s_j := \begin{cases} y_j, & y_j > 0, \\ \mathrm{NA}, & y_j = 0, \end{cases} \tag{4.1}$$ eq.12

where NA *indicates that the realization of S_j is unobserved.*

Let

$$\mathcal{I}_t := \{j \in \{1, \ldots, t\} : y_j > 0\}, \qquad n_t := |\mathcal{I}_t| \tag{4.2}$$ eq.14

denote the set of periods with positive losses and its cardinality. When $n_t \geq 1$, write

$$\mathcal{I}_t = \{i_1, \ldots, i_{n_t}\}, \qquad i_1 < \cdots < i_{n_t},$$

and define the corresponding severity vector and its observed value by

$$S_{\mathcal{I}_t} := (S_{i_1}, \ldots, S_{i_{n_t}})^\top, \qquad s_{\mathcal{I}_t} := (y_{i_1}, \ldots, y_{i_{n_t}})^\top.$$

The observed history $Y_{1:t} = y_{1:t}$ is therefore equivalently represented by

$$Z_{1:t} = z_{1:t} \qquad \textit{and} \qquad S_{\mathcal{I}_t} = s_{\mathcal{I}_t}.$$

When $n_t = 0$, the severity vectors are understood to be empty, and conditioning on the observed history reduces to conditioning on $Z_{1:t} = z_{1:t}$ alone.

thm.2.uni **Theorem 2** (Closed-form one-step mean prediction)**.** *Consider Model 2 with the realized observations*

$$Y_{1:t} = y_{1:t} \in \mathbb{R}^t_{\geq 0},$$

and let

$$Z_{1:t} = z_{1:t}, \qquad S_{\mathcal{I}_t} = s_{\mathcal{I}_t}, \qquad \mathcal{I}_t, \qquad n_t$$

be defined in Remark 1. For the given $Z_{1:t} = z_{1:t}$, define

$$x_1, \ldots, x_t, x_{t+1} \in \{-1, 1\}$$

as in (3.1), and let the $k \times k$ matrix $\boldsymbol{X}_k$ be defined in (3.2) for $k \in \{t, t+1\}$. Then, we have the following results.

i. We have

$$\Theta \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t} \sim N(m_t^{[1]}, C_t^{[1]}), \tag{4.3}$$ eq.posterio

where

$$C_t^{[1]} := \left(\frac{1}{\sigma^2} + \frac{n_t \eta^2}{\tau^2}\right)^{-1} \quad \text{and} \quad m_t^{[1]} := \frac{\eta C_t^{[1]}}{\tau^2} \sum_{j \in \mathcal{I}_t} (\log s_j - d_j). \tag{4.4}$$ eq.222

ii. Furthermore, we have

$$\begin{aligned} &\mathbb{E}\left[Y_{t+1} \mid Y_{1:t} = y_{1:t}\right] \\ &\quad = \exp\left(d_{t+1} + \frac{\tau^2}{2} + \eta m_t^{[1]} + \frac{\eta^2 C_t^{[1]}}{2}\right) \frac{\Phi_{t+1}\left(m_{1:t+1}^{[1**]}; \boldsymbol{I}_{t+1} + C_t^{[1]} \boldsymbol{X}_{t+1}\right)}{\Phi_t\left(m_{1:t}^{[1*]}; \boldsymbol{I}_t + C_t^{[1]} \boldsymbol{X}_t\right)}, \end{aligned}$$

where

$$m_{1:t}^{[1*]} := \left(x_1(c_1 + m_t^{[1]}), \ldots, x_t(c_t + m_t^{[1]})\right)^\top$$

and

$$m_{1:t+1}^{[1**]} := \left(x_1(c_1 + m_t^{[1]} + \eta C_t^{[1]}), \ldots, x_{t+1}(c_{t+1} + m_t^{[1]} + \eta C_t^{[1]})\right)^\top.$$

Proof. We start with the proof of part i. Using

$$\log S_j \mid \Theta \sim N(d_j + \eta\Theta, \tau^2), \qquad j = 1, \ldots, t,$$

and Bayes' formula, the posterior density of Θ at $\theta \in \mathbb{R}$ conditional on $S_{\mathcal{I}_t} = s_{\mathcal{I}_t}$, denoted by

$\pi(\theta \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t})$, is given by

$$\begin{aligned}
&\pi(\theta \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t}) \\
&\quad \propto \pi(\theta) f(s_{\mathcal{I}_t} \mid \Theta = \theta) \\
&\quad \propto \exp\left(-\frac{\theta^2}{2\sigma^2} - \frac{1}{2\tau^2} \sum_{j \in \mathcal{I}_t} (\log s_j - d_j - \eta\theta)^2 \right) \\
&\quad \propto \exp\left(-\frac{1}{2C_t^{[1]}} (\theta - m_t^{[1]})^2 \right),
\end{aligned}$$

where all proportionalities are with respect to θ, and the last proportionality follows from (4.4). This proves part i.

Now, we prove part ii. By the definition of conditional expectation, we have

$$\begin{aligned}
\mathbb{E}\left[Y_{t+1} \mid Y_{1:t} = y_{1:t}\right] &= \mathbb{E}\left[Y_{t+1} \mid Z_{1:t} = z_{1:t}, S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right] \\
&= \frac{\mathbb{E}\left[Y_{t+1} 1_{[Z_{1:t} = z_{1:t}]} \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right]}{\mathbb{P}\left(Z_{1:t} = z_{1:t} \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right)}.
\end{aligned} \tag{4.5}$$

eq.6

We first calculate the denominator

$$\mathbb{P}\left(Z_{1:t} = z_{1:t} \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right).$$

By the conditional independence in Model 2,

$$\begin{aligned}
\mathbb{P}\left(Z_{1:t} = z_{1:t} \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right) &= \mathbb{E}\left[\mathbb{P}\left(Z_{1:t} = z_{1:t} \mid \Theta, S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right) \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right] \\
&= \mathbb{E}\left[\prod_{j=1}^{t} \Phi\left(x_j(c_j + \Theta)\right) \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right] \\
&= \Phi_t\left(m_{1:t}^{[1*]}; \boldsymbol{I}_t + C_t^{[1]} \boldsymbol{X}_t\right),
\end{aligned} \tag{4.6}$$

eq.9

where the last equality follows from Lemma 3 combined with the following observation obtained from part i:

$$\left[(x_1(c_1 + \Theta), \ldots, x_t(c_t + \Theta))^\top \;\middle|\; S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right] \sim \mathrm{MVN}\left(m_{1:t}^{[1*]}, C_t^{[1]} \boldsymbol{X}_t\right).$$

Now, we calculate the numerator

$$\mathbb{E}\left[Y_{t+1} 1_{[Z_{1:t} = z_{1:t}]} \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right]$$

in (4.5). First, observe that

$$\begin{aligned}\mathbb{E}\left[Y_{t+1} \mid \Theta\right] &= \mathbb{E}\left[Z_{t+1} \mid \Theta\right] \mathbb{E}\left[S_{t+1} \mid \Theta\right] \\ &= \Phi(c_{t+1} + \Theta) \exp\left(d_{t+1} + \eta\Theta + \frac{\tau^2}{2}\right).\end{aligned}$$

Then, using this and the tower property, we have

$$\begin{aligned}&\mathbb{E}\left[Y_{t+1} 1_{[Z_{1:t}=z_{1:t}]} \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right] \\ &\quad= \mathbb{E}\left[\mathbb{E}\left[Y_{t+1} \mid \Theta\right] \mathbb{P}\left(Z_{1:t} = z_{1:t} \mid \Theta\right) \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right] \\ &\quad= \exp\left(d_{t+1} + \frac{\tau^2}{2}\right) \times \mathbb{E}\left[\exp(\eta\Theta)\Phi(c_{t+1} + \Theta) \prod_{j=1}^{t} \Phi\left(x_j(c_j + \Theta)\right) \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right] \qquad (4.7) \\ &\quad= \exp\left(d_{t+1} + \frac{\tau^2}{2}\right) \mathbb{E}\left[\exp(\eta\Theta) \prod_{j=1}^{t+1} \Phi\left(x_j(c_j + \Theta)\right) \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right],\end{aligned}$$

eq.numerato

where the last equality follows from the definition $x_{t+1} = 1$.

Since the probit function is finite, applying Lemma 1 with $h = \eta$ to (4.7), we have

$$\begin{aligned}&\mathbb{E}\left[Y_{t+1} 1_{[Z_{1:t}=z_{1:t}]} \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right] \\ &\quad= \exp\left(d_{t+1} + \frac{\tau^2}{2} + \eta m_t^{[1]} + \frac{\eta^2 C_t^{[1]}}{2}\right) \\ &\qquad\times \mathbb{E}\left[\prod_{j=1}^{t+1} \Phi\left(x_j(c_j + \widetilde{\Theta})\right) \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right],\end{aligned} \qquad (4.8)$$

eq.7

where

$$\widetilde{\Theta} \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t} \sim N(m_t^{[1]} + \eta C_t^{[1]}, C_t^{[1]}).$$

Since

$$\begin{aligned}&\left[\left(x_1(c_1 + \widetilde{\Theta}), \ldots, x_{t+1}(c_{t+1} + \widetilde{\Theta})\right)^\top \;\middle|\; S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right] \\ &\quad\sim \mathrm{MVN}\left(m_{1:t+1}^{[1**]}, C_t^{[1]} \boldsymbol{X}_{t+1}\right),\end{aligned}$$

applying Lemma 3 to (4.8), we have

$$\begin{aligned}&\mathbb{E}\left[Y_{t+1} 1_{[Z_{1:t}=z_{1:t}]} \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right] \\ &\quad= \exp\left(d_{t+1} + \frac{\tau^2}{2} + \eta m_t^{[1]} + \frac{\eta^2 C_t^{[1]}}{2}\right) \\ &\qquad\times \Phi_{t+1}\left(m_{1:t+1}^{[1**]}; \boldsymbol{I}_{t+1} + C_t^{[1]} \boldsymbol{X}_{t+1}\right).\end{aligned}$$

Combining this expression with (4.6) in (4.5) concludes the proof of part ii. □

op.loss.uni **Theorem 3** (Closed-form one-step stop-loss prediction)**.** *Under the same setting as in Theorem 2, let $d > 0$. Then, we have*

$$\begin{aligned}&\mathbb{E}\left[(Y_{t+1}-d)_+ \mid Y_{1:t}=y_{1:t}\right]\\&= \frac{\exp\left(d_{t+1}+\frac{\tau^2}{2}+\eta m_t^{[1]}+\frac{\eta^2 C_t^{[1]}}{2}\right)\Phi_{t+2}\left(\mu_{1:t+2}^{[1*]};\boldsymbol{\Omega}_{t+2}^{[1]}\right)-d\,\Phi_{t+2}\left(\mu_{1:t+2}^{[1**]};\boldsymbol{\Omega}_{t+2}^{[1]}\right)}{\Phi_t\left(m_{1:t}^{[1*]};\boldsymbol{I}_t+C_t^{[1]}\boldsymbol{X}_t\right)},\end{aligned} \tag{4.9}$$

eq.stop.loss

where

$$\mu_{1:t+2}^{[1*]} := \begin{pmatrix} m_{1:t+1}^{[1**]} \\ \dfrac{d_{t+1}+\eta m_t^{[1]}+\eta^2 C_t^{[1]}+\tau^2-\log d}{\tau}\end{pmatrix}$$

and

$$\mu_{1:t+2}^{[1**]} := \begin{pmatrix} m_{1:t}^{[1*]} \\ x_{t+1}(c_{t+1}+m_t^{[1]}) \\ \dfrac{d_{t+1}+\eta m_t^{[1]}-\log d}{\tau}\end{pmatrix},$$

with

$$\boldsymbol{\Omega}_{t+2}^{[1]} := \begin{pmatrix} \boldsymbol{I}_{t+1}+C_t^{[1]}\boldsymbol{X}_{t+1} & \dfrac{\eta C_t^{[1]}}{\tau}x_{1:t+1} \\ \dfrac{\eta C_t^{[1]}}{\tau}x_{1:t+1}^\top & 1+\dfrac{\eta^2 C_t^{[1]}}{\tau^2}\end{pmatrix}.$$

Proof. Since $d > 0$, observe that we have

$$(Y_{t+1}-d)_+ = Z_{t+1}(S_{t+1}-d)_+.$$

As a result, the conditional independence in Model 2 and the stop-loss expectation of the normal distribution give

$$\begin{aligned}&\mathbb{E}\left[(Y_{t+1}-d)_+ \mid \Theta\right]\\&= \Phi(c_{t+1}+\Theta)\left[\exp\left(d_{t+1}+\eta\Theta+\frac{\tau^2}{2}\right)\Phi\left(\frac{d_{t+1}+\eta\Theta+\tau^2-\log d}{\tau}\right)-d\,\Phi\left(\frac{d_{t+1}+\eta\Theta-\log d}{\tau}\right)\right].\end{aligned}$$

Similar to (4.5), we have

$$\begin{aligned}&\mathbb{E}\left[(Y_{t+1}-d)_+ \mid Y_{1:t}=y_{1:t}\right]\\&\quad=\frac{\mathbb{E}\left[(Y_{t+1}-d)_+ 1_{[Z_{1:t}=z_{1:t}]} \mid S_{\mathcal{I}_t}=s_{\mathcal{I}_t}\right]}{\mathbb{P}\left(Z_{1:t}=z_{1:t} \mid S_{\mathcal{I}_t}=s_{\mathcal{I}_t}\right)}.\end{aligned} \tag{4.10}$$

eq.stop.loss

The denominator is obtained from (4.6).

We now calculate the numerator. Using the preceding conditional stop-loss expectation, the tower property, the conditional independence in Model 2, and $x_{t+1}=1$, we obtain

$$\begin{aligned}&\mathbb{E}\left[(Y_{t+1}-d)_+ 1_{[Z_{1:t}=z_{1:t}]} \mid S_{\mathcal{I}_t}=s_{\mathcal{I}_t}\right]\\&\quad=\exp\left(d_{t+1}+\frac{\tau^2}{2}\right)\mathbb{E}\left[\exp(\eta\Theta)\prod_{j=1}^{t+1}\Phi\left(x_j(c_j+\Theta)\right)\Phi\left(\frac{d_{t+1}+\eta\Theta+\tau^2-\log d}{\tau}\right)\;\middle|\; S_{\mathcal{I}_t}=s_{\mathcal{I}_t}\right]\\&\qquad-d\,\mathbb{E}\left[\prod_{j=1}^{t+1}\Phi\left(x_j(c_j+\Theta)\right)\Phi\left(\frac{d_{t+1}+\eta\Theta-\log d}{\tau}\right)\;\middle|\; S_{\mathcal{I}_t}=s_{\mathcal{I}_t}\right].\end{aligned} \tag{4.11}$$

eq.stop.loss

For the expectation in the first term in (4.11), since the product of the Gaussian CDFs is bounded by 1 and observing the Gaussian distribution in part i of Theorem 2, Lemma 1 with $h=\eta$ gives.

$$\begin{aligned}&\mathbb{E}\left[\exp(\eta\Theta)\prod_{j=1}^{t+1}\Phi\left(x_j(c_j+\Theta)\right)\Phi\left(\frac{d_{t+1}+\eta\Theta+\tau^2-\log d}{\tau}\right)\;\middle|\; S_{\mathcal{I}_t}=s_{\mathcal{I}_t}\right]\\&\quad=\exp\left(\eta m_t^{[1]}+\frac{\eta^2 C_t^{[1]}}{2}\right)\mathbb{E}\left[\prod_{j=1}^{t+1}\Phi\left(x_j(c_j+\widetilde{\Theta})\right)\Phi\left(\frac{d_{t+1}+\eta\widetilde{\Theta}+\tau^2-\log d}{\tau}\right)\;\middle|\; S_{\mathcal{I}_t}=s_{\mathcal{I}_t}\right],\end{aligned}$$

where

$$\widetilde{\Theta}\mid S_{\mathcal{I}_t}=s_{\mathcal{I}_t}\sim N(m_t^{[1]}+\eta C_t^{[1]}, C_t^{[1]}).$$

Here,

$$\left[\left(\begin{array}{c}x_1(c_1+\widetilde{\Theta})\\\vdots\\x_{t+1}(c_{t+1}+\widetilde{\Theta})\\\dfrac{d_{t+1}+\eta\widetilde{\Theta}+\tau^2-\log d}{\tau}\end{array}\right)\;\middle|\; S_{\mathcal{I}_t}=s_{\mathcal{I}_t}\right]$$

follows the Gaussian distribution with mean $\mu_{1:t+2}^{[1*]}$ and covariance matrix

$$\begin{pmatrix} C_t^{[1]} \boldsymbol{X}_{t+1} & \dfrac{\eta C_t^{[1]}}{\tau} x_{1:t+1} \\ \dfrac{\eta C_t^{[1]}}{\tau} x_{1:t+1}^\top & \dfrac{\eta^2 C_t^{[1]}}{\tau^2} \end{pmatrix} = \boldsymbol{\Omega}_{t+2}^{[1]} - \boldsymbol{I}_{t+2},$$

which follows from the conditional distribution of $\widetilde{\Theta}$.

Therefore, Lemma 3 gives

$$\begin{aligned} &\mathbb{E}\left[\exp(\eta\Theta)\prod_{j=1}^{t+1}\Phi\left(x_j(c_j+\Theta)\right)\Phi\left(\frac{d_{t+1}+\eta\Theta+\tau^2-\log d}{\tau}\right)\;\middle|\; S_{\mathcal{I}_t}=s_{\mathcal{I}_t}\right] \\ &\quad=\exp\left(\eta m_t^{[1]}+\frac{\eta^2 C_t^{[1]}}{2}\right)\Phi_{t+2}\left(\mu_{1:t+2}^{[1*]};\boldsymbol{\Omega}_{t+2}^{[1]}\right). \end{aligned} \tag{4.12}$$

eq.stop.loss

Now, we calculate the expectation in the second term in (4.11). Similar to the distribution in the first term, part i of Theorem 2 shows that

$$\left[\begin{pmatrix} x_1(c_1+\Theta) \\ \vdots \\ x_{t+1}(c_{t+1}+\Theta) \\ \dfrac{d_{t+1}+\eta\Theta-\log d}{\tau} \end{pmatrix}\;\middle|\; S_{\mathcal{I}_t}=s_{\mathcal{I}_t}\right] \sim \mathrm{MVN}_{t+2}\left(\mu_{1:t+2}^{[1**]},\boldsymbol{\Omega}_{t+2}^{[1]}-\boldsymbol{I}_{t+2}\right).$$

Hence, another application of Lemma 3 gives

$$\begin{aligned} &\mathbb{E}\left[\prod_{j=1}^{t+1}\Phi\left(x_j(c_j+\Theta)\right)\Phi\left(\frac{d_{t+1}+\eta\Theta-\log d}{\tau}\right)\;\middle|\; S_{\mathcal{I}_t}=s_{\mathcal{I}_t}\right] \\ &\quad=\Phi_{t+2}\left(\mu_{1:t+2}^{[1**]};\boldsymbol{\Omega}_{t+2}^{[1]}\right). \end{aligned} \tag{4.13}$$

eq.stop.loss

Consequently, substituting (4.12) and (4.13) into (4.11), we obtain

$$\begin{aligned} &\mathbb{E}\left[(Y_{t+1}-d)_+ 1_{[Z_{1:t}=z_{1:t}]}\mid S_{\mathcal{I}_t}=s_{\mathcal{I}_t}\right] \\ &\quad=\exp\left(d_{t+1}+\frac{\tau^2}{2}+\eta m_t^{[1]}+\frac{\eta^2 C_t^{[1]}}{2}\right)\Phi_{t+2}\left(\mu_{1:t+2}^{[1*]};\boldsymbol{\Omega}_{t+2}^{[1]}\right)-d\,\Phi_{t+2}\left(\mu_{1:t+2}^{[1**]};\boldsymbol{\Omega}_{t+2}^{[1]}\right). \end{aligned}$$

Dividing this expression by the denominator obtained from (4.6) proves the result. □

Continuous-discrete mixed nature of Y_t, the likelihood function of $Y_{1:t} = y_{1:t}$ is defined by

$$\ell(y_{1:t}) := \mathbb{P}\left(Z_{1:t} = z_{1:t} \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right) f(s_{\mathcal{I}_t}), \tag{4.14}$$

eq.likeliho

where $f(s_{\mathcal{I}_t})$ denotes the probability mass function of $S_{\mathcal{I}_t}$ evaluated at $s_{\mathcal{I}_t}$. When $|\mathcal{I}_t| = 0$, we set $f(s_{\mathcal{I}_t}) = 1$.

elihood.uni

Corollary 2 (Closed-form likelihood)**.** *Under the same setting as in Theorem 2, the likelihood in (4.14) is given by*

$$\ell(y_{1:t}) = \Phi_t\left(m_{1:t}^{[1*]}; \boldsymbol{I}_t + C_t^{[1]}\boldsymbol{X}_t\right) \frac{\sqrt{C_t^{[1]}}}{\sigma\tau^{n_t}(2\pi)^{n_t/2}\prod_{j\in\mathcal{I}_t} s_j} \exp\left(-\frac{1}{2\tau^2}\sum_{j\in\mathcal{I}_t}(\log s_j - d_j)^2 + \frac{\left(m_t^{[1]}\right)^2}{2C_t^{[1]}}\right). \tag{4.15}$$

eq.likeliho

Proof. From (4.14), it is enough to calculate

$$\mathbb{P}\left(Z_{1:t} = z_{1:t} \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right) \quad \text{and} \quad f(s_{\mathcal{I}_t}).$$

First, the conditional probability can be obtained from (4.6).

It remains to calculate $f(s_{\mathcal{I}_t})$. Suppose that $n_t = |\mathcal{I}_t| > 0$. Under Model 2, we have

$$\Theta \sim N(0, \sigma^2), \qquad S_j \mid \Theta \sim \mathrm{LN}(d_j + \eta\Theta, \tau^2), \qquad j \in \mathcal{I}_t.$$

Using the conditional independence of the severities given Θ, we obtain

$$\begin{aligned}
f(s_{\mathcal{I}_t}) &= \int_{\mathbb{R}} \frac{1}{\sigma}\phi\left(\frac{\theta}{\sigma}\right)\prod_{j\in\mathcal{I}_t} f(s_j \mid \Theta = \theta)\,\mathrm{d}\theta \\
&= \frac{1}{\sigma\tau^{n_t}(2\pi)^{(n_t+1)/2}\prod_{j\in\mathcal{I}_t} s_j}\int_{\mathbb{R}} \exp\left(-\frac{\theta^2}{2\sigma^2} - \frac{1}{2\tau^2}\sum_{j\in\mathcal{I}_t}(\log s_j - d_j - \eta\theta)^2\right)\mathrm{d}\theta \\
&= \frac{1}{\sigma\tau^{n_t}(2\pi)^{(n_t+1)/2}\prod_{j\in\mathcal{I}_t} s_j}\int_{\mathbb{R}} \exp\left(-\frac{(\theta - m_t^{[1]})^2}{2C_t^{[1]}} - \frac{1}{2\tau^2}\sum_{j\in\mathcal{I}_t}(\log s_j - d_j)^2 + \frac{\left(m_t^{[1]}\right)^2}{2C_t^{[1]}}\right)\mathrm{d}\theta \\
&= \frac{\sqrt{C_t^{[1]}}}{\sigma\tau^{n_t}(2\pi)^{n_t/2}\prod_{j\in\mathcal{I}_t} s_j}\exp\left(-\frac{1}{2\tau^2}\sum_{j\in\mathcal{I}_t}(\log s_j - d_j)^2 + \frac{\left(m_t^{[1]}\right)^2}{2C_t^{[1]}}\right),
\end{aligned}$$

where the third equality follows from (4.4). When $n_t = 0$, we have $C_t^{[1]} = \sigma^2$ and $m_t^{[2]} = 0$, so the

same expression gives $f(s_{\mathcal{I}_t}) = 1$, in accordance with the definition.

Combining this expression with (4.6) in (4.14) proves the result. □

sterior.uni **Corollary 3** (Posterior distribution of the random effect)**.** *Under the setting of Theorem 2, the conditional distribution*

$$\Theta \mid Y_{1:t} = y_{1:t}$$

has density $\pi(\theta \mid Y_{1:t} = y_{1:t})$ *at* $\theta \in \mathbb{R}$ *given by*

$$\pi(\theta \mid Y_{1:t} = y_{1:t}) = \frac{\dfrac{1}{\sqrt{C_t^{[1]}}}\phi\left(\dfrac{\theta - m_t^{[1]}}{\sqrt{C_t^{[1]}}}\right)\displaystyle\prod_{j=1}^{t}\Phi\left(x_j(c_j+\theta)\right)}{\Phi_t\left(m_{1:t}^{[1*]}; \boldsymbol{I}_t + C_t^{[1]}\boldsymbol{X}_t\right)}. \tag{4.16}$$ eq.posterio

Proof. Using Remark 1 and the conditional independence in Model 2, Bayes' formula gives

$$\begin{aligned}\pi(\theta \mid Y_{1:t} = y_{1:t}) &= \frac{\pi(\theta \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t})\mathbb{P}\left(Z_{1:t} = z_{1:t} \mid \Theta = \theta\right)}{\mathbb{P}\left(Z_{1:t} = z_{1:t} \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right)} \\ &= \frac{\pi(\theta \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t})\displaystyle\prod_{j=1}^{t}\Phi\left(x_j(c_j+\theta)\right)}{\mathbb{P}\left(Z_{1:t} = z_{1:t} \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right)}.\end{aligned}$$

By part i of Theorem 2, we have

$$\pi(\theta \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t}) = \frac{1}{\sqrt{C_t^{[1]}}}\phi\left(\frac{\theta - m_t^{[1]}}{\sqrt{C_t^{[1]}}}\right).$$

Substituting this expression and (4.6) into Bayes' formula gives (4.16), which proves the result. □

5. Extension to bivariate random effect

model.2 **Model 3** (Bernoulli-lognormal probit bivariate random effect model)**.** *Let*

$$\rho \in [-1, 1], \quad \sigma_1, \sigma_2, \tau > 0, \quad (c_t)_{t\in\mathbb{N}}, (d_t)_{t\in\mathbb{N}} \subseteq \mathbb{R}$$

be predetermined exogenous constants. We extend Model 2 by replacing the shared random effect with a bivariate random effect as follows.

i. Random effect: Assume

$$\Theta_{1:2} \sim \mathrm{MVN}_2\left(0_{1:2}, \begin{pmatrix}\sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2\end{pmatrix}\right).$$

ii. Observation equations: Retain the conditional independence assumptions of Model 2, now conditional on $\Theta_{1:2}$, and replace the observation equations by

$$Z_t \mid \Theta_{1:2} \sim \mathrm{Ber}\left(\Phi(c_t + \Theta_1)\right), \qquad S_t \mid \Theta_{1:2} \sim \mathrm{LN}(d_t + \Theta_2, \tau^2).$$

As before, the observed loss is defined by

$$Y_t = Z_t S_t, \qquad t \in \mathbb{N}.$$

thm.biv **Theorem 4** (Closed-form one-step mean prediction)**.** *Consider Model 3, with $\rho \in (-1, 1)$, and the realized observations*

$$Y_{1:t} = y_{1:t} \in \mathbb{R}^t_{\geq 0},$$

and let

$$Z_{1:t} = z_{1:t}, \qquad S_{\mathcal{I}_t} = s_{\mathcal{I}_t}, \qquad \mathcal{I}_t, \qquad n_t$$

be defined in Remark 1. For this given $Z_{1:t} = z_{1:t}$, define

$$x_1, \ldots, x_t, x_{t+1} \in \{-1, 1\}$$

as in (3.1), and let the $k \times k$ matrix $\boldsymbol{X}_k$ be defined in (3.2) for $k \in \{t, t+1\}$. Then, we have the following results.

i. We have

$$\Theta_{1:2} \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t} \sim \mathrm{MVN}_2\left(\begin{pmatrix} \frac{\rho\sigma_1}{\sigma_2} m_t^{[2]} \\ m_t^{[2]} \end{pmatrix}, \begin{pmatrix} \sigma_1^2(1-\rho^2) + \frac{\rho^2\sigma_1^2}{\sigma_2^2} C_t^{[2]} & \frac{\rho\sigma_1}{\sigma_2} C_t^{[2]} \\ \frac{\rho\sigma_1}{\sigma_2} C_t^{[2]} & C_t^{[2]} \end{pmatrix}\right), \tag{5.1}$$ eq2.1

where

$$C_t^{[2]} := \left(\frac{1}{\sigma_2^2} + \frac{n_t}{\tau^2}\right)^{-1} \quad \text{and} \quad m_t^{[2]} := \frac{C_t^{[2]}}{\tau^2} \sum_{j \in \mathcal{I}_t} (\log s_j - d_j). \tag{5.2}$$ eq.22

ii. Furthermore, we have

$$\mathbb{E}\left[Y_{t+1} \mid Y_{1:t} = y_{1:t}\right] = \exp\left(d_{t+1} + \frac{\tau^2}{2} + m_t^{[2]} + \frac{C_t^{[2]}}{2}\right) \frac{\Phi_{t+1}\left(m_{1:t+1}^{[2**]}; \boldsymbol{I}_{t+1} + \left\{\sigma_1^2(1-\rho^2) + \frac{\rho^2\sigma_1^2}{\sigma_2^2} C_t^{[2]}\right\} \boldsymbol{X}_{t+1}\right)}{\Phi_t\left(m_{1:t}^{[1*]}; \boldsymbol{I}_t + \left\{\sigma_1^2(1-\rho^2) + \frac{\rho^2\sigma_1^2}{\sigma_2^2} C_t^{[2]}\right\} \boldsymbol{X}_t\right)}.$$

where

$$m_{1:t}^{[2*]} := \left(x_1 \left(c_1 + \frac{\rho\sigma_1}{\sigma_2} m_t^{[2]} \right), \cdots, x_t \left(c_t + \frac{\rho\sigma_1}{\sigma_2} m_t^{[2]} \right) \right)^\top$$

and

$$m_{1:t+1}^{[2**]} := \left(x_1 \left(c_1 + \frac{\rho\sigma_1}{\sigma_2} (m_t^{[2]} + C_t^{[2]}) \right), \cdots, x_{t+1} \left(c_{t+1} + \frac{\rho\sigma_1}{\sigma_2} (m_t^{[2]} + C_t^{[2]}) \right) \right)^\top$$

Proof. First, observe that the same calculation as in part i of Theorem 2 gives

$$\Theta_2 \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t} \sim N(m_t^{[2]}, C_t^{[2]}).$$

Moreover, the conditional independence of Θ_1 and $S_{\mathcal{I}_t}$ given Θ_2 implies

$$[\Theta_1 \mid \Theta_2, S_{\mathcal{I}_t} = s_{\mathcal{I}_t}] \sim N\left(\frac{\rho\sigma_1}{\sigma_2} \Theta_2, \sigma_1^2(1-\rho^2) \right).$$

Combining these two conditional Gaussian distributions gives (5.1), which proves part i.

Now, we prove part ii. As in the proof of Theorem 2, we have

$$\mathbb{E}\left[Y_{t+1} \mid Y_{1:t} = y_{1:t} \right] = \frac{\mathbb{E}\left[Y_{t+1} 1_{[Z_{1:t} = z_{1:t}]} \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t} \right]}{\mathbb{P}\left(Z_{1:t} = z_{1:t} \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t} \right)}. \tag{5.3}$$ eq.condition

We first calculate the denominator. By part i,

$$[\Theta_1 \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t}] \sim N\left(\frac{\rho\sigma_1}{\sigma_2} m_t^{[2]}, \sigma_1^2(1-\rho^2) + \frac{\rho^2\sigma_1^2}{\sigma_2^2} C_t^{[2]} \right). \tag{5.4}$$ eq.24

Hence, the conditional independence in Model 3 and Lemma 3 give

$$\begin{aligned} \mathbb{P}\left(Z_{1:t} = z_{1:t} \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t} \right) &= \mathbb{E}\left[\prod_{j=1}^{t} \Phi\left(x_j (c_j + \Theta_1) \right) \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t} \right] \\ &= \Phi_t\left(m_{1:t}^{[2*]}; \boldsymbol{I}_t + \left(\sigma_1^2(1-\rho^2) + \frac{\rho^2\sigma_1^2}{\sigma_2^2} C_t^{[2]} \right) \boldsymbol{X}_t \right) \end{aligned} \tag{5.5}$$ eq.denomina

where the last equality follows from

$$\left[(x_1(c_1 + \Theta_1), \ldots, x_t(c_t + \Theta_1))^\top \;\middle|\; S_{\mathcal{I}_t} = s_{\mathcal{I}_t} \right] \sim \mathrm{MVN}\left(m_{1:t}^{[2*]}, \left(\sigma_1^2(1-\rho^2) + \frac{\rho^2\sigma_1^2}{\sigma_2^2} C_t^{[2]} \right) \boldsymbol{X}_t \right).$$

Now, we calculate the numerator. Using the conditional independence in Model 3 and $x_{t+1} = 1$,

the same argument as in (4.7) gives

$$\begin{aligned}&\mathbb{E}\left[Y_{t+1}1_{[Z_{1:t}=z_{1:t}]} \mid S_{\mathcal{I}_t}=s_{\mathcal{I}_t}\right]\\&\quad=\exp\left(d_{t+1}+\frac{\tau^2}{2}\right)\mathbb{E}\left[\exp(\Theta_2)\prod_{j=1}^{t+1}\Phi\left(x_j(c_j+\Theta_1)\right) \mid S_{\mathcal{I}_t}=s_{\mathcal{I}_t}\right].\end{aligned} \tag{5.6}$$

eq.numerato

Since the product of the Gaussian CDFs is bounded by 1, Lemma 2 with $h_{1:2}=(0,1)^\top$ to the conditional expectation in (5.6), we have

$$\mathbb{E}\left[\exp(\Theta_2)\prod_{j=1}^{t+1}\Phi\left(x_j(c_j+\Theta_1)\right) \mid S_{\mathcal{I}_t}=s_{\mathcal{I}_t}\right]=\exp\left(m_t^{[2]}+\frac{C_t^{[2]}}{2}\right)\mathbb{E}\left[\prod_{j=1}^{t+1}\Phi\left(x_j(c_j+\widetilde{\Theta}_1)\right) \mid S_{\mathcal{I}_t}=s_{\mathcal{I}_t}\right],$$

where

$$\left[\widetilde{\Theta}_{1:2} \mid S_{\mathcal{I}_t}=s_{\mathcal{I}_t}\right]\sim \mathrm{MVN}_2\left(\begin{pmatrix}\frac{\rho\sigma_1}{\sigma_2}(m_t^{[2]}+C_t^{[2]})\\ m_t^{[2]}+C_t^{[2]}\end{pmatrix},\begin{pmatrix}\sigma_1^2(1-\rho^2)+\frac{\rho^2\sigma_1^2}{\sigma_2^2}C_t^{[2]} & \frac{\rho\sigma_1}{\sigma_2}C_t^{[2]}\\ \frac{\rho\sigma_1}{\sigma_2}C_t^{[2]} & C_t^{[2]}\end{pmatrix}\right). \tag{5.7}$$

eq.23

Applying Lemma 3, we obtain

$$\begin{aligned}&\mathbb{E}\left[Y_{t+1}1_{[Z_{1:t}=z_{1:t}]} \mid S_{\mathcal{I}_t}=s_{\mathcal{I}_t}\right]\\&\quad=\exp\left(d_{t+1}+\frac{\tau^2}{2}+m_t^{[2]}+\frac{C_t^{[2]}}{2}\right)\Phi_{t+1}\left(m_{1:t+1}^{[2**]};\boldsymbol{I}_{t+1}+\left(\sigma_1^2(1-\rho^2)+\frac{\rho^2\sigma_1^2}{\sigma_2^2}C_t^{[2]}\right)\boldsymbol{X}_{t+1}\right).\end{aligned} \tag{5.8}$$

eq.tilting.l

Combining this expression with (5.5) in (5.3) concludes the proof of part ii. □

op.loss.biv

Theorem 5 (Closed-form one-step stop-loss prediction)**.** *Under the same setting as in Theorem 4, let $d>0$, we have*

$$\begin{aligned}&\mathbb{E}\left[(Y_{t+1}-d)_+ \mid Y_{1:t}=y_{1:t}\right]\\&\quad=\frac{\exp\left(d_{t+1}+\frac{\tau^2}{2}+m_t^{[2]}+\frac{C_t^{[2]}}{2}\right)\Phi_{t+2}\left(\mu_{1:t+2}^{[2*]};\boldsymbol{\Omega}_{t+2}^{[2]}\right)-d\,\Phi_{t+2}\left(\mu_{1:t+2}^{[2**]};\boldsymbol{\Omega}_{t+2}^{[2]}\right)}{\Phi_t\left(m_{1:t}^{[2*]};\boldsymbol{I}_t+\left(\sigma_1^2(1-\rho^2)+\frac{\rho^2\sigma_1^2}{\sigma_2^2}C_t^{[2]}\right)\boldsymbol{X}_t\right)}\end{aligned}$$

where

$$\mu_{1:t+2}^{[2*]} := \begin{pmatrix} m_{1:t+1}^{[2**]} \\ \dfrac{d_{t+1} + m_t^{[2]} + C_t^{[2]} + \tau^2 - \log d}{\tau} \end{pmatrix} \quad \textit{and} \quad \mu_{1:t+2}^{[2**]} := \begin{pmatrix} m_{1:t}^{[2*]} \\ x_{t+1}\left(c_{t+1} + \dfrac{\rho\sigma_1}{\sigma_2} m_t^{[2]}\right) \\ \dfrac{d_{t+1} + m_t^{[2]} - \log d}{\tau} \end{pmatrix}$$

and

$$\boldsymbol{\Omega}_{t+2}^{[2]} := \begin{pmatrix} \boldsymbol{I}_{t+1} + \left(\sigma_1^2(1-\rho^2) + \dfrac{\rho^2\sigma_1^2}{\sigma_2^2} C_t^{[2]}\right) \boldsymbol{X}_{t+1} & \dfrac{\rho\sigma_1 C_t^{[2]}}{\sigma_2\tau} x_{1:t+1} \\ \dfrac{\rho\sigma_1 C_t^{[2]}}{\sigma_2\tau} x_{1:t+1}^\top & 1 + \dfrac{C_t^{[2]}}{\tau^2} \end{pmatrix}.$$

Proof. As in the proof of Theorem 3, we have

$$\mathbb{E}\left[(Y_{t+1} - d)_+ \mid Y_{1:t} = y_{1:t}\right] = \frac{\mathbb{E}\left[(Y_{t+1} - d)_+ 1_{[Z_{1:t} = z_{1:t}]} \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right]}{\mathbb{P}\left(Z_{1:t} = z_{1:t} \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right)}.$$

The denominator is obtained from (5.5).

We now calculate the numerator. The same conditional stop-loss calculation as in (4.11), using the conditional independence in Model 3 and $x_{t+1} = 1$, gives

$$\begin{aligned} &\mathbb{E}\left[(Y_{t+1} - d)_+ 1_{[Z_{1:t} = z_{1:t}]} \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right] \\ &\quad = \exp\left(d_{t+1} + \frac{\tau^2}{2}\right) \mathbb{E}\left[\exp(\Theta_2) \prod_{j=1}^{t+1} \Phi\left(x_j(c_j + \Theta_1)\right) \Phi\left(\frac{d_{t+1} + \Theta_2 + \tau^2 - \log d}{\tau}\right) \,\middle|\, S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right] \\ &\qquad - d\,\mathbb{E}\left[\prod_{j=1}^{t+1} \Phi\left(x_j(c_j + \Theta_1)\right) \Phi\left(\frac{d_{t+1} + \Theta_2 - \log d}{\tau}\right) \,\middle|\, S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right]. \end{aligned} \tag{5.9}$$

eq.25

For the expectation in the first term, the product of the Gaussian CDFs is bounded by 1, Lemma 2 with $h_{1:2} = (0,1)^\top$, followed by Lemma 3, we obtain

$$\begin{aligned} &\mathbb{E}\left[\exp(\Theta_2) \prod_{j=1}^{t+1} \Phi\left(x_j(c_j + \Theta_1)\right) \Phi\left(\frac{d_{t+1} + \Theta_2 + \tau^2 - \log d}{\tau}\right) \,\middle|\, S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right] \\ &\quad = \exp\left(m_t^{[2]} + \frac{C_t^{[2]}}{2}\right) \mathbb{E}\left[\prod_{j=1}^{t+1} \Phi\left(x_j(c_j + \widetilde{\Theta}_1)\right) \Phi\left(\frac{d_{t+1} + \widetilde{\Theta}_2 + \tau^2 - \log d}{\tau}\right) \,\middle|\, S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right] \\ &\quad = \exp\left(m_t^{[2]} + \frac{C_t^{[2]}}{2}\right) \Phi_{t+2}\left(\mu_{1:t+2}^{[2*]}; \boldsymbol{\Omega}_{t+2}^{[2]}\right). \end{aligned} \tag{5.10}$$

eq.26

Here, the conditional distribution of $\widetilde{\Theta}_{1:2}$ is given by (5.7), and the last equality follows from the following observation

$$\left[\left(\begin{array}{c} x_1(c_1+\widetilde{\Theta}_1) \\ \vdots \\ x_{t+1}(c_{t+1}+\widetilde{\Theta}_1) \\ \underline{d_{t+1}+\widetilde{\Theta}_2+\tau^2-\log d} \\ \tau \end{array}\right) \,\middle|\, S_{\mathcal{I}_t}=s_{\mathcal{I}_t}\right] \sim \mathrm{MVN}_{t+2}\left(\mu_{1:t+2}^{[2*]}, \boldsymbol{\Omega}_{t+2}^{[2]}-\boldsymbol{I}_{t+2}\right).$$

For the expectation in the second term in (5.9), part i of Theorem 4 gives

$$\left[\left(\begin{array}{c} x_1(c_1+\Theta_1) \\ \vdots \\ x_{t+1}(c_{t+1}+\Theta_1) \\ \underline{d_{t+1}+\Theta_2-\log d} \\ \tau \end{array}\right) \,\middle|\, S_{\mathcal{I}_t}=s_{\mathcal{I}_t}\right] \sim \mathrm{MVN}_{t+2}\left(\mu_{1:t+2}^{[2**]}, \boldsymbol{\Omega}_{t+2}^{[2]}-\boldsymbol{I}_{t+2}\right).$$

Hence, applying Lemma 3, we have

$$\mathbb{E}\left[\prod_{j=1}^{t+1}\Phi\left(x_j(c_j+\Theta_1)\right)\Phi\left(\frac{d_{t+1}+\Theta_2-\log d}{\tau}\right) \,\middle|\, S_{\mathcal{I}_t}=s_{\mathcal{I}_t}\right] = \Phi_{t+2}\left(\mu_{1:t+2}^{[2**]}; \boldsymbol{\Omega}_{t+2}^{[2]}\right). \tag{5.11}$$

eq.27

Substituting (5.10) and (5.11) into (5.9) and dividing by the denominator in (5.5) proves the result. □

cor2.1 **Corollary 4** (Closed-form likelihood)**.** *Under the same setting as in Theorem 4, the likelihood in* (4.14) *is given by*

$$\begin{aligned}\ell(y_{1:t}) = \Phi_t\left(m_{1:t}^{[2*]}; \boldsymbol{I}_t + \left\{\sigma_1^2(1-\rho^2)+\frac{\rho^2\sigma_1^2}{\sigma_2^2}C_t^{[2]}\right\}\boldsymbol{X}_t\right) \\ \times \frac{\sqrt{C_t^{[2]}}}{\sigma_2\tau^{n_t}(2\pi)^{n_t/2}\prod_{j\in\mathcal{I}_t}s_j}\exp\left\{-\frac{1}{2\tau^2}\sum_{j\in\mathcal{I}_t}(\log s_j-d_j)^2+\frac{\left(m_t^{[2]}\right)^2}{2C_t^{[2]}}\right\}.\end{aligned}$$

Proof. The conditional probability in (4.14) is given by (5.5). It remains to calculate $f(s_{\mathcal{I}_t})$.

Since the conditional distribution of the severities depends on the random effect only through

$\Theta_2 \sim N(0, \sigma_2^2)$, the similar calculation as in the proof of Corollary 2 for $n_t > 0$ gives

$$
\begin{aligned}
f(s_{\mathcal{I}_t}) &= \frac{1}{\sigma_2 \tau^{n_t} (2\pi)^{(n_t+1)/2} \prod_{j \in \mathcal{I}_t} s_j} \int_{\mathbb{R}} \exp\left(-\frac{\theta_2^2}{2\sigma_2^2} - \frac{1}{2\tau^2} \sum_{j \in \mathcal{I}_t} (\log s_j - d_j - \theta_2)^2 \right) \mathrm{d}\theta_2 \\
&= \frac{\sqrt{C_t^{[2]}}}{\sigma_2 \tau^{n_t} (2\pi)^{n_t/2} \prod_{j \in \mathcal{I}_t} s_j} \exp\left(-\frac{1}{2\tau^2} \sum_{j \in \mathcal{I}_t} (\log s_j - d_j)^2 + \frac{\left(m_t^{[2]}\right)^2}{2C_t^{[2]}} \right),
\end{aligned}
$$

where the last equality follows by completing the square using (5.2). When $n_t = 0$, we have $C_t^{[2]} = \sigma_2^2$ and $m_t^{[2]} = 0$, so the this expression gives $f(s_{\mathcal{I}_t}) = 1$, in accordance with the definition. Combining this expression with (5.5) in (4.14) proves the result. □

cor.61 **Corollary 5** (Posterior distribution of the random effect)**.** *Under the setting of Theorem 4, the conditional distribution* $\Theta_{1:2} \mid Y_{1:t} = y_{1:t}$ *has density* $\pi(\theta_{1:2} \mid Y_{1:t} = y_{1:t})$ *at* $\theta_{1:2} = (\theta_1, \theta_2)^\top \in \mathbb{R}^2$ *given by*

$$
\begin{aligned}
&\pi(\theta_{1:2} \mid Y_{1:t} = y_{1:t}) \\
&\quad = \frac{\phi_2\left(\begin{pmatrix} \theta_1 - \dfrac{\rho\sigma_1}{\sigma_2} m_t^{[2]} \\ \theta_2 - m_t^{[2]} \end{pmatrix} ; \begin{pmatrix} \sigma_1^2(1-\rho^2) + \dfrac{\rho^2\sigma_1^2}{\sigma_2^2} C_t^{[2]} & \dfrac{\rho\sigma_1}{\sigma_2} C_t^{[2]} \\ \dfrac{\rho\sigma_1}{\sigma_2} C_t^{[2]} & C_t^{[2]} \end{pmatrix} \right) \prod_{j=1}^{t} \Phi\left(x_j(c_j + \theta_1)\right)}{\Phi_t\left(m_{1:t}^{[2*]} ; \boldsymbol{I}_t + \left(\sigma_1^2(1-\rho^2) + \dfrac{\rho^2\sigma_1^2}{\sigma_2^2} C_t^{[2]} \right) \boldsymbol{X}_t \right)}.
\end{aligned}
\tag{5.12}
$$

eq.posterio

Proof. As in the proof of Corollary 3, Bayes' formula and the conditional independence in Model 3 give

$$
\begin{aligned}
&\pi(\theta_{1:2} \mid Y_{1:t} = y_{1:t}) \\
&\quad = \frac{\pi(\theta_{1:2} \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t}) \prod_{j=1}^{t} \Phi\left(x_j(c_j + \theta_1)\right)}{\mathbb{P}\left(Z_{1:t} = z_{1:t} \mid S_{\mathcal{I}_t} = s_{\mathcal{I}_t}\right)}.
\end{aligned}
$$

Substituting the Gaussian density from part i of Theorem 4 and the conditional probability in (5.5) gives (5.12), which proves the result. □

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References

2022scalable Cao, J., Durante, D., and Genton, M. G. (2022). Scalable computation of predictive probabilities in probit models with gaussian process priors. *Journal of Computational and Graphical Statistics*, 31(3):709–720.

19conjugate Durante, D. (2019). Conjugate bayes for probit regression via unified skew-normal distributions. *Biometrika*, 106(4):765–779.

well2017two Farewell, V., Long, D., Tom, B., Yiu, S., and Su, L. (2017). Two-part and related regression models for longitudinal data. *Annual review of statistics and its application*, 4:283–315.

lo2021closed Fasano, A., Rebaudo, G., Durante, D., and Petrone, S. (2021). A closed-form filter for binary time series. *Statistics and Computing*, 31(4):47.

10dependent Frees, E. W. J., Meyers, G., and Cummings, A. D. (2010). Dependent multi-peril ratemaking models. *ASTIN Bulletin: The Journal of the IAA*, 40(2):699–726.

uo2025entity Guo, J., Quan, Z., and Zhang, L. (2025). Entity-specific cyber risk assessment using insurtech empowered risk factors. *arXiv preprint arXiv:2507.08193*.

oh2026matrix Oh, M., Jeong, H., Kim, D., and Jung, K. (2026). Matrix-based factor analysis on the prediction of insurance claims probability. *ASTIN Bulletin: The Journal of the IAA*, pages 1–18.

yiu2018two Yiu, S. and Tom, B. D. (2018). Two-part models with stochastic processes for modelling longitudinal semicontinuous data: Computationally efficient inference and modelling the overall marginal mean. *Statistical methods in medical research*, 27(12):3679–3695.

ultivariate Young, G., Valdez, E. A., and Kohn, R. (2009). Multivariate probit models for conditional claim-types. *Insurance: Mathematics and Economics*, 44(2):214–228.

Appendix A. Miscelleaneous results

app.1